

Unit 5. Variable Selection and Dimension Reduction

1 Subset Selection

FREES §5.2, §6.2–6.3 · ISLR §6.1.1–6.1.2.

Definition 5.1 (Search procedures). With k candidate predictors:

- (i) **Best subset selection** fits all 2^k models and keeps, for each size d , the one with smallest SSE.
- (ii) **Forward stepwise** starts from the intercept-only model and at each step adds the predictor giving the largest reduction in SSE.
- (iii) **Backward stepwise** starts from the full model and at each step deletes the predictor giving the smallest increase in SSE.

Proposition 5.2 (Cost of the search). Best subset fits 2^k models; each stepwise procedure fits $1 + k(k + 1)/2$.

Proof. There are $\binom{k}{d}$ subsets of size d and $\sum_d \binom{k}{d} = 2^k$. Forward stepwise fits one null model, then k models at the first step, $k - 1$ at the second, and so on, giving $1 + \sum_{j=1}^k j = 1 + k(k + 1)/2$;

backward stepwise is the same count in reverse. □

Remark. At $k = 20$ the two counts are 1,048,576 and 211. The saving is not free: stepwise is greedy and can miss the best subset of a given size, since a variable useless alone may be essential in company. Backward stepwise additionally requires $n > k$, so it is unavailable in high dimensions, whereas forward stepwise is not.

2 Selection Criteria

FREES §5.6 · ISLR §6.1.3.

Write d for the number of estimated coefficients including the intercept, $\mathbf{H}$ for the projection onto the fitted model's column space, $\mathbf{M} = \mathbf{I} - \mathbf{H}$, and let $\hat{\sigma}^2$ estimate σ^2 from a model believed adequate, usually the full one.

2.1 The optimism of the training error

Definition 5.3 (In-sample prediction error). Let $\mathbf{Y} = \boldsymbol{\mu} + \boldsymbol{\varepsilon}$ with $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, and let $\mathbf{Y}^{\text{new}} = \boldsymbol{\mu} + \boldsymbol{\varepsilon}^{\text{new}}$ be an independent replicate at the same design. The **in-sample prediction error** is

$$\text{Err}_{\text{in}} = \frac{1}{n} \mathbb{E} |\mathbf{Y}^{\text{new}} - \hat{\mathbf{Y}}|^2, \quad \hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}. \quad (1)$$

Theorem 5.4 (Mallows). For any $\boldsymbol{\mu}$, whether or not it lies in the fitted model's column space,

$$n \text{Err}_{\text{in}} = \mathbb{E}[\text{SSE}] + 2d\sigma^2. \quad (2)$$

Proof. Both sides are computed from Lemma “expectation of a quadratic form” of the prerequisites, $\mathbb{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \text{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}'_z \mathbf{A} \boldsymbol{\mu}_z$.

The training error. $\text{SSE} = |\mathbf{M}\mathbf{Y}|^2 = \mathbf{Y}'\mathbf{M}\mathbf{Y}$ with $\mathbb{E}[\mathbf{Y}] = \boldsymbol{\mu}$ and $\text{var}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, so

$$\mathbb{E}[\text{SSE}] = \sigma^2 \text{tr}(\mathbf{M}) + \boldsymbol{\mu}'\mathbf{M}\boldsymbol{\mu} = \sigma^2(n - d) + |\mathbf{M}\boldsymbol{\mu}|^2,$$

using $\text{tr}(\mathbf{M}) = n - d$ and $\mathbf{M}'\mathbf{M} = \mathbf{M}$.

The prediction error. Since $\boldsymbol{\varepsilon}^{\text{new}}$ is independent of $\mathbf{Y}$ with mean zero,

$$\mathbb{E}|\mathbf{Y}^{\text{new}} - \hat{\mathbf{Y}}|^2 = \mathbb{E}|\boldsymbol{\mu} - \mathbf{H}\mathbf{Y}|^2 + \mathbb{E}|\boldsymbol{\varepsilon}^{\text{new}}|^2 = \mathbb{E}|\mathbf{M}\boldsymbol{\mu} - \mathbf{H}\boldsymbol{\varepsilon}|^2 + n\sigma^2,$$

where $\boldsymbol{\mu} - \mathbf{H}\mathbf{Y} = \boldsymbol{\mu} - \mathbf{H}\boldsymbol{\mu} - \mathbf{H}\boldsymbol{\varepsilon} = \mathbf{M}\boldsymbol{\mu} - \mathbf{H}\boldsymbol{\varepsilon}$. The cross term vanishes because $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$, and $\mathbb{E}|\mathbf{H}\boldsymbol{\varepsilon}|^2 = \sigma^2 \text{tr}(\mathbf{H}) = \sigma^2 d$, so

$$n \text{Err}_{\text{in}} = |\mathbf{M}\boldsymbol{\mu}|^2 + \sigma^2 d + n\sigma^2.$$

Subtracting, $n \text{Err}_{\text{in}} - \mathbb{E}[\text{SSE}] = \sigma^2 d - \sigma^2(n - d) + n\sigma^2 = 2d\sigma^2$. □

Remark. The bias term $|\mathbf{M}\boldsymbol{\mu}|^2$ cancels, so (2) holds under misspecification. The training error understates prediction error by $2d\sigma^2/n$, and that quantity grows with the number of parameters, which is the precise form of Unit 1's optimism of the training error.

Definition 5.5 (Criteria).

$$C_p = \frac{1}{n}(\text{SSE} + 2d \hat{\sigma}^2), \quad (3)$$

$$\text{AIC} = \frac{1}{n}(\text{SSE} + 2d \hat{\sigma}^2), \quad (4)$$

$$\text{BIC} = \frac{1}{n}(\text{SSE} + \ln(n) d \hat{\sigma}^2), \quad (5)$$

$$\bar{R}^2 = 1 - \frac{\text{SSE}/(n-d)}{\text{SST}/(n-1)}. \quad (6)$$

Smaller is better for the first three, larger for the fourth.

Corollary 5.6. C_p with $\hat{\sigma}^2 = \sigma^2$ is unbiased for Err_{in} .

Proof. Divide (2) by n and compare with (3). □

Proposition 5.7 (AIC and C_p coincide here). For the Gaussian linear model with σ^2 known, $-2\ell + 2d$ equals $n C_p / \sigma^2$ up to an additive constant free of the model.

Proof. By the log-likelihood of Unit 2, $\ell = -\frac{n}{2} \ln(2\pi\sigma^2) - \text{SSE}/(2\sigma^2)$, so

$$-2\ell + 2d = n \ln(2\pi\sigma^2) + \frac{\text{SSE}}{\sigma^2} + 2d = \frac{1}{\sigma^2} (\text{SSE} + 2d\sigma^2) + n \ln(2\pi\sigma^2).$$

The first term is $n C_p / \sigma^2$ and the second does not depend on which predictors are used. $\square$

Remark. BIC replaces the penalty 2 by $\ln n$, which exceeds 2 once $n > 7$, so BIC selects models at least as small as AIC. The two answer different questions: BIC is consistent for the true model when one is among the candidates, while AIC and C_p target prediction error and do not become model-selection consistent as n grows. Neither dominates.

2.2 Adjusted R^2

Lemma 5.8 (Effect of one added regressor). Let $\mathbf{M}_k$ be the residual maker for $\mathbf{X}_k$ and let $\mathbf{x} \notin \mathcal{C}(\mathbf{X}_k)$ be appended. Then the residual maker of the enlarged design is $\mathbf{M}_{k+1} = \mathbf{M}_k - \mathbf{M}_k \mathbf{x} (\mathbf{x}' \mathbf{M}_k \mathbf{x})^{-1} \mathbf{x}' \mathbf{M}_k$, the new coefficient is $\hat{\beta} = \mathbf{x}' \mathbf{M}_k \mathbf{Y} / (\mathbf{x}' \mathbf{M}_k \mathbf{x})$, and

$$\text{SSE}_k - \text{SSE}_{k+1} = \frac{(\mathbf{x}' \mathbf{M}_k \mathbf{Y})^2}{\mathbf{x}' \mathbf{M}_k \mathbf{x}} = \hat{\beta}^2 \mathbf{x}' \mathbf{M}_k \mathbf{x}. \quad (7)$$

Proof. $\mathcal{C}([\mathbf{X}_k \ \mathbf{x}])$ is spanned by $\mathcal{C}(\mathbf{X}_k)$ together with $\mathbf{M}_k \mathbf{x}$, and these are orthogonal, so the projection onto the enlarged space is the sum of the projection onto $\mathcal{C}(\mathbf{X}_k)$ and the projection onto the line through $\mathbf{M}_k \mathbf{x}$. Subtracting from $\mathbf{I}$,

$$\mathbf{M}_{k+1} = \mathbf{M}_k - \frac{\mathbf{M}_k \mathbf{x} \mathbf{x}' \mathbf{M}_k}{\mathbf{x}' \mathbf{M}_k \mathbf{x}}.$$

Hence $\text{SSE}_{k+1} = \mathbf{Y}' \mathbf{M}_{k+1} \mathbf{Y} = \text{SSE}_k - (\mathbf{x}' \mathbf{M}_k \mathbf{Y})^2 / (\mathbf{x}' \mathbf{M}_k \mathbf{x})$. The coefficient formula is Frisch–Waugh–Lovell (Unit 3) with $\mathbf{X}_1 = \mathbf{x}$, and substituting it gives the right-hand form. $\square$

Theorem 5.9 ($\bar{R}^2$ rises exactly when $|t| > 1$). Adding one regressor increases $\bar{R}^2$ if and only if its t statistic in the enlarged model exceeds 1 in absolute value.

Proof. SST does not depend on the regressors, so $\bar{R}^2$ increases exactly when MSE decreases, that is when

$$\frac{\text{SSE}_{k+1}}{n - d - 1} < \frac{\text{SSE}_k}{n - d},$$

writing d for the number of parameters before the addition. Substituting $\text{SSE}_k = \text{SSE}_{k+1} + \hat{\beta}^2 \mathbf{x}'\mathbf{M}_k\mathbf{x}$ from Lemma 5.8 and clearing denominators,

$$(n - d)\text{SSE}_{k+1} < (n - d - 1)\text{SSE}_{k+1} + (n - d - 1)\hat{\beta}^2 \mathbf{x}'\mathbf{M}_k\mathbf{x},$$

that is $\text{SSE}_{k+1} < (n - d - 1)\hat{\beta}^2 \mathbf{x}'\mathbf{M}_k\mathbf{x}$. Dividing by $\text{SSE}_{k+1} = (n - d - 1)\text{MSE}_{k+1}$ gives

$$1 < \frac{\hat{\beta}^2 \mathbf{x}'\mathbf{M}_k\mathbf{x}}{\text{MSE}_{k+1}}.$$

By Lemma 5.8 and Frisch–Waugh–Lovell, $\text{var}(\hat{\beta}) = \sigma^2/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$, so $\text{SE}(\hat{\beta})^2 = \text{MSE}_{k+1}/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$ and the right-hand side is exactly $t^2 = \hat{\beta}^2/\text{SE}(\hat{\beta})^2$. $\square$

Remark. The usual significance threshold is $|t| > 2$, so $\bar{R}^2$ admits variables that a t test would reject. It penalises size, but weakly; C_p , AIC and BIC penalise more.

Proposition 5.10 (Likelihood-ratio test). For nested Gaussian linear models differing in q constraints, $-2(\ell_R - \ell_U) = n \ln(\text{SSE}_R/\text{SSE}_U)$, which is asymptotically χ_q^2 under the null.

Proof. Maximising the Gaussian likelihood over σ^2 gives $\hat{\sigma}^2 = \text{SSE}/n$ and profile log-likelihood $\ell = -\frac{n}{2} \ln(2\pi\text{SSE}/n) - \frac{n}{2}$, by Unit 2. Differencing the two models, the constants cancel and $-2(\ell_R - \ell_U) = n \ln \text{SSE}_R - n \ln \text{SSE}_U$. The χ_q^2 limit is the general likelihood-ratio result; for these models the exact F test of Unit 3 is available and preferable in small samples. $\square$

3 Dimension Reduction

ISLR §6.3, §12.2.

Dimension reduction replaces the k predictors by $M < k$ linear combinations and regresses on those.

3.1 Principal components

Let $\mathbf{X}$ be $n \times k$ with columns centred, and let $\mathbf{S} = \mathbf{X}'\mathbf{X}/(n-1)$ be the sample covariance matrix, symmetric and positive semi-definite.

Definition 5.11 (Principal components). The **first principal component loading vector** is a unit $\phi_1 \in \mathbb{R}^k$ maximising the sample variance of the scores $\mathbf{X}\phi$, namely $\phi'\mathbf{S}\phi$. The j -th loading vector maximises the same quantity subject to being orthogonal to $\phi_1, \dots, \phi_{j-1}$, and $\mathbf{z}_j = \mathbf{X}\phi_j$ is the j -th **score** vector.

Theorem 5.12 (Principal components are eigenvectors). Let $\mathbf{S}$ have eigenvalues $\lambda_1 \geq \dots \geq \lambda_k \geq 0$ with orthonormal eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_k$. Then $\phi_j = \mathbf{v}_j$ and the maximised variance is λ_j . The score vectors are uncorrelated, with $\mathbf{z}_j'\mathbf{z}_l/(n-1) = \lambda_j\delta_{jl}$.

Proof. The spectral theorem of the prerequisites gives the eigendecomposition, and its Rayleigh corollary gives $\max_{|\phi|=1} \phi'\mathbf{S}\phi = \lambda_1$, attained at $\mathbf{v}_1$: that is the case $j = 1$.

For $j > 1$, write $\phi = \sum_l c_l \mathbf{v}_l$ with $\sum_l c_l^2 = 1$. Orthogonality to $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$ forces $c_1 = \dots = c_{j-1} = 0$, so $\phi'\mathbf{S}\phi = \sum_{l \geq j} \lambda_l c_l^2 \leq \lambda_j$, with equality at $\phi = \mathbf{v}_j$.

For the scores, $\mathbf{z}_j' \mathbf{z}_l = \boldsymbol{\phi}_j' \mathbf{X}' \mathbf{X} \boldsymbol{\phi}_l = (n-1) \mathbf{v}_j' \mathbf{S} \mathbf{v}_l = (n-1) \lambda_l \mathbf{v}_j' \mathbf{v}_l$, which is $(n-1) \lambda_j$ when $j = l$ and zero otherwise. $\square$

Definition 5.13 (Proportion of variance explained). The j -th component explains the fraction $\text{PVE}_j = \lambda_j / \sum_l \lambda_l$ of the total sample variance.

Proposition 5.14. $\sum_l \lambda_l = \text{tr}(\mathbf{S}) = \sum_j s_{jj}$, so the PVE_j sum to 1.

Proof. $\text{tr}(\mathbf{S}) = \text{tr}(\mathbf{V} \boldsymbol{\Lambda} \mathbf{V}') = \text{tr}(\boldsymbol{\Lambda})$ by the cyclic property of the trace (prerequisites), and $\text{tr}(\boldsymbol{\Lambda}) = \sum_l \lambda_l$. $\square$

Remark (Scaling is a modelling choice). $\mathbf{S}$ changes under rescaling of the columns, and so do its eigenvectors: a variable measured in metres rather than kilometres has 10^6 times the variance and will dominate the first component. Standardising each column to unit variance replaces $\mathbf{S}$ by the correlation matrix and removes the dependence on units. It is the default unless the variables are already in comparable units.

3.2 Principal components regression

Definition 5.15 (PCR). **Principal components regression** with M components regresses $\mathbf{Y}$ on $\mathbf{z}_1, \dots, \mathbf{z}_M$ and an intercept.

Proposition 5.16 (PCR with all components is least squares). If $\mathbf{X}$ has full column rank and $M = k$, the PCR fit equals the least squares fit of $\mathbf{Y}$ on $\mathbf{X}$.

Proof. The score matrix is $\mathbf{Z} = \mathbf{XV}$ with $\mathbf{V}$ orthogonal, hence invertible, so $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X})$. Least squares fitted values are the orthogonal projection onto the column space (Unit 2), and the two column spaces coincide. $\square$

Remark. So PCR is not a new estimator but a path from the null model to least squares, indexed by M . It reduces variance by discarding the low-variance directions of $\mathbf{X}$, at the cost of bias if the response happens to depend on them. The directions are chosen without looking at $\mathbf{Y}$, so PCR can discard exactly the direction that matters.

3.3 Partial least squares

Definition 5.17 (PLS). The first **partial least squares** direction is ϕ_1 with $\phi_{1j} \propto \langle \mathbf{x}_j, \mathbf{Y} \rangle$, the coefficient from regressing $\mathbf{Y}$ on $\mathbf{x}_j$ alone. Subsequent directions repeat the construction after replacing each $\mathbf{x}_j$ by its residual from regressing on the directions already found.

Remark. PLS uses $\mathbf{Y}$ in choosing the directions, so it targets directions that both vary and predict, whereas PCA targets variance alone. In practice the gain over PCR is usually small: the extra variance introduced by estimating the directions from $\mathbf{Y}$ offsets the bias reduction.

4 Computation: Formulas Against Package Output

ISLR §6.5.

Data: **Hitters** from ISLR2, salary on the other variables, rows with missing salary removed.

```
1 suppressMessages({library(ISLR2); library(leaps); library(pls)})
2 d <- na.omit(Hitters); n <- nrow(d)
3 full <- lm(Salary ~ ., data = d); k <- length(coef(full)) - 1L
4 s2 <- sum(residuals(full)^2) / (n - k - 1)      # sigma^2 from the full model
```

```
5 c(n = n, k = k, sigma2 = s2)
```

```
##      n      k sigma2
##    263     19 99591
```

4.1 The criteria (§2)

```
1 m <- lm(Salary ~ AtBat + Hits + Walks + CRBI + Division + PutOuts, data = d)
2 dd <- length(coef(m)); RSS <- sum(residuals(m)^2)
3 SST <- sum((d$Salary - mean(d$Salary))^2)
4 c(Cp      = (RSS + 2*dd*s2)/n,
5   AIC     = (RSS + 2*dd*s2)/n,           # same formula, ISLR 6.1.3
6   BIC     = (RSS + log(n)*dd*s2)/n,
7   adjR2_manual = 1 - (RSS/(n-dd)) / (SST/(n-1)),
8   adjR2_lm    = summary(m)$adj.r.squared)
```

```
##      Cp      AIC      BIC adjR2_manual
## 104901.8362 104901.8362 114370.6181      0.4972
##      adjR2_lm
##      0.4972
```

```
1 set.seed(1)
```

```

2 X <- model.matrix(full); nn <- nrow(X)
3 Hm <- X %*% solve(t(X) %*% X) %*% t(X); dof <- ncol(X)
4 mu <- drop(X %*% coef(full)); sg <- 200; B <- 3000
5 res <- replicate(B, {
6   y <- mu + rnorm(nn, sd = sg)
7   yh <- drop(Hm %*% y)
8   c(SSE = sum((y - yh)^2),
9     pred = sum((mu + rnorm(nn, sd = sg) - yh)^2))
10 })
11 c(E_SSE = mean(res["SSE", ]), n_Err_in = mean(res["pred", ]),
12   difference = mean(res["pred", ]) - mean(res["SSE", ]),
13   two_d_sigma2 = 2 * dof * sg^2)

```

```
##           E_SSE      n_Err_in  difference two_d_sigma2
##      9749364      11323890      1574525      1600000
```

4.2 Subset selection (§1)

```

1 rs <- regsubsets(Salary ~ ., data = d, nvmax = 19)
2 sm <- summary(rs)
3 c(best_by_Cp = which.min(sm$cp), best_by_BIC = which.min(sm$bic),

```

```
4 best_by_adjR2 = which.max(sm$adjr2))
```

```
##      best_by_Cp    best_by_BIC best_by_adjR2
##              10              6             11
```

The three criteria disagree, and in the predicted direction: BIC is the most parsimonious, $\bar{R}^2$ the least, with C_p between.

```
1 c(best_subset_models = 2^k, stepwise_models = 1 + k*(k+1)/2) # Proposition 5.2
```

```
## best_subset_models    stepwise_models
##              524288              191
```

```
1 fw <- summary(regsubsets(Salary ~ ., data = d, nvmax = 19, method = "forward"))
2 same <- sapply(1:19, function(j) identical(sm$which[j, ], fw$which[j, ]))
3 c(sizes_where_forward_matches_best = sum(same), of = 19,
4   first_disagreement = which(!same)[1])
```

```
## sizes_where_forward_matches_best
##              17
##              of
##              19
```

```
##                first_disagreement
##                                7
```

Greedy search is not exact: forward stepwise reproduces the best subset at some sizes and not others.

4.3 Principal components (§3)

```
1 Xs <- scale(model.matrix(full)[, -1])           # centred and scaled
2 S  <- cov(Xs)
3 ev <- eigen(S); pc <- prcomp(Xs, center = FALSE, scale. = FALSE)
4 c(lambda1_eigen = ev$values[1], pc_sdev2 = pc$sdev[1]^2,   # Theorem 5.13
5    loading_max_diff = max(abs(abs(ev$vectors[, 1]) - abs(pc$rotation[, 1]))),
6    scores_uncorrelated = max(abs(cor(pc$x)[upper.tri(diag(ncol(Xs))))]))
```

```
##          lambda1_eigen          pc_sdev2    loading_max_diff
##          7.280e+00          7.280e+00          4.441e-16
## scores_uncorrelated
##          4.890e-15
```

```
1 c(sum_lambda = sum(ev$values), trace_S = sum(diag(S)),      # Proposition 5.14
2   PVE1 = ev$values[1]/sum(ev$values),
3   PVE_sums_to_one = sum(ev$values/sum(ev$values)))
```

```
##          sum_lambda          trace_S          PVE1
##          19.0000          19.0000          0.3831
## PVE_sums_to_one
##          1.0000
```

```
1 pcrfit <- pcr(Salary ~ ., data = d, scale = TRUE, validation = "none")
2 c(max_diff_vs_OLS = max(abs(predict(pcrfit, ncomp = 19) - fitted(full))))
```

```
## max_diff_vs_OLS
##          5.457e-12
```

PCR with all 19 components reproduces least squares to machine precision, which is Proposition 5.16.

5 Pitfalls

5.1 Reporting the selected model's p -values

The t statistics of a model chosen by searching over subsets are not t distributed: the selection used the same data. Below, no predictor has any effect.

```
1 set.seed(2)
2 nS <- 100; pS <- 40
3 Xn <- matrix(rnorm(nS*pS), nS, pS); yn <- rnorm(nS)
```

```

4 sel <- order(abs(cor(Xn, yn)), decreasing = TRUE)[1:5]
5 ms  <- lm(yn ~ Xn[, sel])
6 c(n_significant_at_5pct = sum(coef(summary(ms))[-1, 4] < 0.05),
7   smallest_p = min(coef(summary(ms))[-1, 4]),
8   R2 = summary(ms)$r.squared)

```

```

## n_significant_at_5pct          smallest_p
##                1.00000                0.04191
##                R2
##                0.13940

```

Diagnostic. Significant coefficients in a model whose variables were chosen for their association with the response. **Fix.** Hold out data not used in the selection, or cross-validate the whole procedure as in Unit 1.

5.2 Choosing $\hat{\sigma}^2$ from the model being scored

C_p requires an estimate of σ^2 from a model believed adequate, and substituting the candidate's own estimate makes the criterion degenerate. In the standardised form $C_p = \text{RSS}_d / \hat{\sigma}^2 - n + 2(d + 1)$,

putting $\hat{\sigma}^2 = \text{RSS}_d / (n - d - 1)$ gives $\text{RSS}_d / \hat{\sigma}^2 = n - d - 1$ identically, so

$$C_p = (n - d - 1) - n + 2(d + 1) = d + 1,$$

a function of the size alone: the data have dropped out and the smallest model always wins. The residual sums of squares are extracted through the model matrix, since **regsubsets** names the dummy columns of a factor and those are not columns of the data frame.

```
1 Xd <- model.matrix(Salary ~ ., d); sizes <- 1:19
2 RSS <- sapply(sizes, function(j)
3   sum(residuals(lm(d$Salary ~ Xd[, names(coef(rs, j))[-1], drop = FALSE)))^2))
4 std_full <- RSS/s2 - n + 2*(sizes + 1)
5 std_own  <- RSS/(RSS/(n - sizes - 1)) - n + 2*(sizes + 1)
6 rbind(size = sizes, sigma2_from_full = round(std_full, 2),
7       sigma2_from_own = round(std_own, 2),
8       d_plus_1 = sizes + 1)[, c(1:6, 18:19)]
```

```
##           [,1]  [,2]  [,3]  [,4]  [,5]  [,6]  [,7]
## size           1.0  2.00  3.00  4.00  5.00  6.00 18.00
## sigma2_from_full 104.3 50.72 38.69 27.86 21.61 14.02 18.01
## sigma2_from_own   2.0  3.00  4.00  5.00  6.00  7.00 19.00
```

```
## d_plus_1      2.0  3.00  4.00  5.00  6.00  7.00 19.00
##              [,8]
## size          19
## sigma2_from_full 20
## sigma2_from_own  20
## d_plus_1      20
```

```
1 c(chosen_from_full = which.min(std_full), chosen_from_own = which.min(std_own),
2   largest_gap_to_d_plus_1 = max(abs(std_own - (sizes + 1))))
```

```
##          chosen_from_full      chosen_from_own
##          1.000e+01          1.000e+00
## largest_gap_to_d_plus_1
##          2.842e-14
```

Diagnostic. A C_p column equal to $d + 1$, or more generally one that is a function of the model size alone and moves not at all when the data change. **Fix.** Estimate σ^2 once, from the full model or another adequate one. The scale-free form used elsewhere in this unit, $C_p = \{\text{RSS}_d + 2(d + 1)\hat{\sigma}^2\}/n$, does not collapse in the same way: the same substitution turns it into $(\text{RSS}_d/n)(n + d + 1)/(n - d - 1)$, which still penalises size. It is nonetheless a different criterion from the one intended, and on these

data it is not the collapse but the silent change of criterion that is the hazard.

5.3 Standardisation before PCA

Principal components are not scale invariant.

```
1 Xr <- model.matrix(full)[, -1]
2 Xr2 <- Xr; Xr2[, "CRuns"] <- Xr2[, "CRuns"] * 1000      # a change of units
3 c(PVE1_unscaled_original = summary(prcomp(Xr))$importance[2, 1],
4   PVE1_unscaled_reunited = summary(prcomp(Xr2))$importance[2, 1],
5   PVE1_scaled_original   = summary(prcomp(Xr, scale. = TRUE))$importance[2, 1],
6   PVE1_scaled_reunited   = summary(prcomp(Xr2, scale. = TRUE))$importance[2, 1])
```

```
## PVE1_unscaled_original PVE1_unscaled_reunited
##                0.9744                1.0000
## PVE1_scaled_original   PVE1_scaled_reunited
##                0.3831                0.3831
```

Diagnostic. A first component dominated by whichever variable has the largest numerical spread.

Fix. Scale to unit variance unless the variables are already comparable.

6 Exercises

Exercise 5.1. With $k = 15$ candidate predictors, how many models does best subset selection fit, and how many does forward stepwise? At which size do they first differ in the number of models considered?

Solution. By Proposition 5.2, $2^{15} = 32,768$ and $1 + 15 \cdot 16/2 = 121$.

They already differ at size 1 in the sense of coverage, but not in count: both examine all 15 single-predictor models. At size 2 best subset examines $\binom{15}{2} = 105$ models while forward stepwise examines only the 14 that extend its chosen first variable. That is where the greedy restriction first bites. ■

Exercise 5.2. A model with $d = 6$ parameters fitted to $n = 100$ observations has $\text{SSE} = 2400$, and $\hat{\sigma}^2 = 27$ from the full model. Compute C_p and BIC. A second model has $d = 9$ and $\text{SSE} = 2250$; which is preferred by each criterion?

Solution. For the first, $C_p = (2400 + 2 \cdot 6 \cdot 27)/100 = (2400 + 324)/100 = 27.24$ and $\text{BIC} = (2400 + \ln(100) \cdot 6 \cdot 27)/100 = (2400 + 746.0)/100 = 31.46$, using $\ln 100 = 4.605$.

For the second, $C_p = (2250 + 2 \cdot 9 \cdot 27)/100 = (2250 + 486)/100 = 27.36$ and $\text{BIC} = (2250 + 4.605 \cdot 9 \cdot 27)/100 = (2250 + 1119.0)/100 = 33.69$.

C_p prefers the smaller model, $27.24 < 27.36$, and BIC prefers it more decisively, $31.46 < 33.69$. The extra three parameters bought a SSE reduction of 150, less than the 162 that C_p charges for them. ■

Exercise 5.3. Prove that $\mathbf{E}[\text{SSE}] + 2d\sigma^2 = n \text{Err}_{\text{in}}$ and explain why the result does not require the fitted model to be correct.

Solution. The proof is Theorem 5.4. The bias term is $|\mathbf{M}\boldsymbol{\mu}|^2$, which measures how far the true mean lies from the fitted model's column space; it appears in $\mathbf{E}[\text{SSE}]$ and in $n \text{Err}_{\text{in}}$ with the same coefficient and therefore cancels on subtraction. Nothing in the argument assumed $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$: only $\mathbf{E}[\mathbf{Y}] = \boldsymbol{\mu}$, $\text{var}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, and that $\mathbf{H}$ is a projection of rank d . ■

Exercise 5.4. Show that adding a regressor raises $\bar{R}^2$ exactly when $|t| > 1$, and say what this implies about using $\bar{R}^2$ to select.

Solution. The proof is Theorem 5.9. The threshold $|t| > 1$ is far weaker than the conventional $|t| > 2$, so $\bar{R}^2$ retains variables that a significance test would discard, and it therefore selects models that are systematically too large. It penalises size, but not enough to be a serious criterion; C_p , AIC and BIC all charge more. ■

Exercise 5.5. For what n does BIC penalise more heavily than AIC? What does each converge to as n grows, when the true model is among the candidates?

Solution. The penalties are $2d\hat{\sigma}^2$ and $\ln(n)d\hat{\sigma}^2$, so BIC is the heavier as soon as $\ln n > 2$, that is $n > e^2 = 7.39$, so from $n = 8$ onward.

As $n \rightarrow \infty$, BIC selects the true model with probability tending to one, being model-selection consistent. AIC does not: its penalty is too light, and it retains spurious variables with probability bounded away from zero. In exchange AIC and C_p target prediction error, and when no candidate is correct that is the more relevant target. ■

Exercise 5.6. Show that the first principal component loading vector is the leading eigenvector of the sample covariance matrix, and that the score vectors are uncorrelated.

Solution. Both are Theorem 5.12. The first claim is the Rayleigh characterisation of the largest eigenvalue applied to $\mathbf{S}$, which is symmetric, so the spectral theorem provides an orthonormal eigenbasis. The second follows from $\mathbf{z}'_j \mathbf{z}_l = (n-1)\lambda_l \mathbf{v}'_j \mathbf{v}_l$, which vanishes for $j \neq l$ by orthonormality of the eigenvectors. Uncorrelatedness makes the regression on the scores in PCR a sequence of independent one-dimensional problems. ■

Exercise 5.7. PCR with $M < k$ components and best subset selection with M predictors both produce a model using M derived quantities. How do they differ, and when would each fail?

Solution. Subset selection keeps M of the original predictors and discards the rest, so the fitted model is interpretable in the original variables. PCR keeps M linear combinations of **all** predictors, so no variable is discarded and interpretation is in terms of components.

Subset selection fails when the signal is spread thinly across many correlated predictors, none individually strong. PCR fails when the response depends on a low-variance direction of $\mathbf{X}$, since the directions are chosen from $\mathbf{X}$ alone without reference to $\mathbf{Y}$; that direction is discarded early however predictive it is. Partial least squares addresses that particular failure by using $\mathbf{Y}$ in the construction. ■

Exercise 5.8. Why does standardising the columns change the principal components, and when should it be omitted?

Solution. The components maximise $\phi' \mathbf{S} \phi$, and $\mathbf{S}$ is not invariant under rescaling of the columns: multiplying a column by c multiplies its variance by c^2 and its covariances by c . A variable recorded in smaller units acquires a larger variance and pulls the first component toward itself, regardless of any substantive importance.

Standardisation should be omitted when the variables are already in the same units and their relative variances are meaningful, for instance repeated measurements of one quantity at several times, where rescaling each separately would destroy information. ■

Exercise 5.9. A colleague fits all 2^{20} subsets, reports the model with the smallest SSE at its size, and quotes its t statistics. What is wrong, and what would you report instead?

Solution. The model was chosen because it fits this data set best, so its coefficients are the extreme order statistics of a large search and their t statistics are inflated. The reference distribution for a t statistic assumes the variable was fixed in advance, which it was not, so the quoted p -values

are not p -values (§5.1).

Report instead an error estimate computed on data untouched by the selection, or cross-validate the entire procedure, selection included, as in Unit 1. If inference on the coefficients is really wanted, the honest route is to select on one half of the data and fit on the other. ■

Exercise 5.10. In R: (a) verify Theorem 5.4 by simulation; (b) confirm that PCR with all components equals least squares; (c) show that the principal components change when one column is rescaled but not when the columns are standardised.

Solution. (a) is §4.1, (b) §4.3 and (c) §5.3. In (a) note that the simulated difference between the two expectations should match $2d\sigma^2$ up to Monte Carlo error, and that the agreement does not depend on the fitted model being correct. ■